Exp 7

Server

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <pthread.h>
#define PORT 8080
#define MAX_CLIENTS 10
#define BUFFER_SIZE 1024
int clients[MAX_CLIENTS];
int client_count = 0;
pthread_mutex_t lock;
void broadcast(char *message, int sender)
{
    pthread_mutex_lock(&lock);
    for (int i = 0; i < client_count; i++)
    {
        if (clients[i] != sender)
        {
            send(clients[i], message, strlen(message), 0);
        }
    }
    pthread_mutex_unlock(&lock);
}
void *handle_client(void *arg)
{
    int client_socket = *((int *)arg);
    char buffer[BUFFER_SIZE];
    int len;
    while ((len = recv(client_socket, buffer, BUFFER_SIZE - 1, 0)) > 0)
    {
        buffer[len] = '\0';
        broadcast(buffer, client_socket);
    }
    close(client_socket);
    pthread_exit(NULL);
}
```

```
int main()
{
    int server_fd, client_socket;
    struct sockaddr_in server_addr, client_addr;
    socklen_t addr_size;
    pthread_mutex_init(&lock, NULL);
    server_fd = socket(AF_INET, SOCK_STREAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(PORT);
    server_addr.sin_addr.s_addr = INADDR_ANY;
    bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    listen(server_fd, 5);
    printf("Server running on port %d\n", PORT);
    while (1)
    {
        addr_size = sizeof(client_addr);
        client_socket = accept(server_fd, (struct sockaddr *)&client_addr, &addr_size);
        clients[client_count++] = client_socket;
        pthread_t tid;
        pthread_create(&tid, NULL, handle_client, &clients[client_count - 1]);
    }
    close(server_fd);
    return 0;
}
```

Client

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <pthread.h>
#define PORT 8080
#define BUFFER_SIZE 1024
int sockfd;
void *receive_msg(void *arg)
{
    char message[BUFFER_SIZE];
    int len;
```

```
    while ((len = recv(sockfd, message, BUFFER_SIZE - 1, 0)) > 0)
    {
        message[len] = '\0';
        printf("%s", message);
    }
    return NULL;
}
int main()
{
    struct sockaddr_in server_addr;
    char message[BUFFER_SIZE];
    sockfd = socket(AF_INET, SOCK_STREAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(PORT);
    inet_pton(AF_INET, "127.0.0.1", &server_addr.sin_addr);
    connect(sockfd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    pthread_t tid;
    pthread_create(&tid, NULL, receive_msg, NULL);
    while (fgets(message, BUFFER_SIZE, stdin))
    {
        send(sockfd, message, strlen(message), 0);
    }
    close(sockfd);
    return 0;
}
```

Server

```
Server running on port 8080
Client connected
Client connected
Received from client: hello
Received from client: how are you
Received from client: i am okky!
```

Client 1

```
hello
Received: how are you
i am okky!
```

Client2

```
Received: hello
how are you
Received: i am okky!
```